

Nuclear-Charge-Guided Mamba with KAN Dynamic Mixture for Molecular Property Prediction

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Article

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Posted Date: December 30th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-8308135/v1>

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Additional Declarations: There is **NO** Competing Interest.

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Abstract

Molecular property prediction (MPP) is a critical yet challenging task in drug discovery, where existing methods face fundamental limitations: Graph Neural Networks (GNNs) suffer from over-smoothing and limited receptive fields, while Graph Transformers incur prohibitive quadratic complexity. To address these challenges, we propose **KAN-NC-Mamba**, a novel molecular representation learning framework that integrates three key innovations. (1) We introduce a **Nuclear-Charge-guided node ordering**, which serializes molecular graphs by ascending atomic number through a chemistry-driven approach to prioritize chemically salient atoms without additional parameters. (2) We design a **Nuclear-Charge-guided Mamba module** that employs selective state-space models to establish global long-range dependencies with linear complexity, effectively overcoming the limited receptive fields of traditional graph neural networks. (3) We develop a **KAN Dynamic Mixture module** based on Kolmogorov-Arnold Networks to achieve nonlinear adaptive fusion of local and global features, breaking through the limitations of traditional linear weighting or simple concatenation fusion paradigms. Experimental results on ten benchmark datasets demonstrate that **KAN-NC-Mamba** achieves state-of-the-art performance on both classification and regression tasks. Ablation studies further validate the effectiveness and complementary nature of each innovative module, showing that the nuclear-charge ordering strategy provides significant performance improvement across multiple tasks compared to popular degree-based ordering. The proposed **KAN-NC-Mamba** not only advances the accuracy-efficiency frontier in molecular machine learning but also provides chemically interpretable representations, as validated by our visualization studies.

Keywords: molecular property prediction, graph representation learning, nuclear-charge ordering, state-space models, Kolmogorov-Arnold networks

1 Introduction

Bringing a single new chemical entity to market now exceeds USD 2.6 billion and 10–15 years, with early-stage *in-silico* screening accounting for $\approx 30\%$ of both cost and timeline [1, 2]. Representative molecular property prediction (MPP) models further

highlight this trend [3–6]. MPP acts as the first “virtual gatekeeper” in this funnel: every 1% improvement in Pearson or ROC-AUC translates into a 3–5% reduction in late-stage attrition, worth hundreds of millions of dollars [7]. Similar observations are echoed in modern MPP benchmarks. Consequently, the central question is not whether we need better MPP models, but *what kind of molecular representation can simultaneously deliver chemical interpretability, linear complexity, and multi-task generalization*.

Two dominant families have emerged, yet both hit a structural ceiling. Sequence-based views encode molecules as SMILES strings and apply RNN or Transformer backbones. While self-attention captures long-range dependencies in principle, the quadratic $\mathcal{O}(L^2)$ cost limits batch size for heavy molecules ($L > 200$). More importantly, chemically adjacent atoms can be separated by tens of characters in SMILES, so functional-group interactions are diluted before the first gradient step [8]. Recent sequence-oriented state-space models offer partial mitigation [9, 10]. Graph-based views use message-passing neural networks (MPNNs) that respect covalent topology; however, the receptive field grows exponentially with depth, leading to over-squashing and over-smoothing [11, 12]. Existing remedies—virtual global nodes or hierarchical pooling—add hyper-parameters and still scale super-linearly [13, 14]. A shared deficiency is the fusion stage: almost all prior work combines local node features with a global graph read-out via static learned weights or simple concatenation [15–17], thereby treating “noise from distant aliphatic chains” and “signal from an ortho-nitrobenzene alert” equally.

Recent advances in structured state-space sequence models (Mamba) offer $\mathcal{O}(L)$ complexity and a global receptive field [18]; see also related molecular and graph-oriented variants. However, they have two blind spots when ported to molecules: (i) node ordering is arbitrary, so chemically important atoms may lie outside the effective window; (ii) the architecture lacks an explicit inductive bias for rings or multi-path topology. Simultaneously, Kolmogorov–Arnold Networks (KANs) have been shown to approximate high-dimensional functions with fewer parameters than ReLU-MLPs via learnable activation functions [19] and subsequent extensions [20, 21], yet their utility for dynamic feature gating remains unexplored.

To address these challenges, we abandon the incremental strategies of “patching Transformers” or “simply deepening GNNs” and return to first principles to reconsider how molecules should be ordered and fused in a computational graph. We propose the **KAN Dynamic Mixture of Nuclear-Charge-Guided Mamba (KAN-NC-Mamba)**. The core insight is to (1) use atomic number Z , an intrinsic descriptor that encodes atomic size, electronegativity, and reactivity, as a universal criterion for node ordering. By serializing the molecular graph into a sequence that is monotonic with respect to Z , chemically critical atoms are positioned within the immediate global context without increasing computational cost. (2) A linear-complexity Mamba state-space model then captures long-range dependencies, circumventing the quadratic bottleneck of self-attention. (3) To overcome the long-standing limitation of static local–global feature fusion, we introduce Kolmogorov–Arnold Networks (KAN) into molecular representation learning for the first time, enabling adaptive, nonlinear mixture weights that are learned in a data-driven manner. Our main contributions are as follows.

- **KAN-NC-Mamba architecture.** We propose a molecular representation framework that couples linear-time Mamba scanning with a KAN-driven dynamic mixture layer, yielding accuracy on par with Graph-Transformers while avoiding quadratic complexity. The KAN gate adaptively fuses local node signals with global graph context through learnable, non-linear basis functions, overcoming the limitation of fixed or linear fusion strategies.
- **Nuclear-charge-based node ordering.** Atoms are sorted by ascending atomic number Z , a fundamental property correlated with size, electronegativity, and reactivity, before sequence encoding. This ordering exposes chemically central atoms to the global receptive field earlier, aligning the serialized sequence with chemical intuition and improving long-range dependency modeling without extra parameters.
- **KAN dynamic mixture module.** Leveraging the Kolmogorov–Arnold representation theorem, the module generates data-dependent mixture weights between local and global features via parametric splines. Compared with traditional additive attention or static weighting, KAN achieves higher expressive power per parameter, granting the model flexibility across diverse molecular scales.
- **Extensive empirical validation.** KAN-NC-Mamba is evaluated on ten widely adopted benchmarks covering physicochemical, bioactivity, and ADMET properties. The model establishes new state-of-the-art results on 9/10 datasets, demonstrating the effectiveness of the proposed ordering strategy and KAN-based fusion mechanism. The visualization results confirm that KAN-NC-Mamba effectively captures chemically meaningful representations for molecular property predictions.

2 Results

2.1 Description of KAN-NC-Mamba

We propose **KAN-NC-Mamba**, a framework for efficient and expressive molecular representation learning through four coordinated components (Fig. 1). The **local message-passing module** (Fig. 1a) uses a Message Passing Neural Network (MPNN) that aggregates neighborhood information along chemical bonds to generate local atom- and bond-level encodings, denoted $\mathbf{H}_{\text{loc}}$ and $\mathbf{E}_{\text{loc}}$, respectively. The **nuclear-charge-guided Mamba module** (Fig. 1b) sorts atoms by their nuclear charge values (Z) and processes the resulting sequence via the NC-Mamba block. This module performs a linear-time selective scan to capture long-range dependencies throughout the molecule, generating global contextual representations $\mathbf{H}_{\text{glob}}$. The **KAN Dynamic Mixture (KDM) module** (Fig. 1c) integrates local and global features in a node-wise manner. By leveraging Kolmogorov–Arnold spline functions, it learns adaptive, nonlinear mixture weights to fuse $\mathbf{H}_{\text{loc}}$ and $\mathbf{H}_{\text{glob}}$, producing a unified graph-level embedding $\mathbf{H}_{\mathcal{G}}$. Finally, the **prediction module** (Fig. 1d) applies an edge-aware readout to $\mathbf{H}_{\mathcal{G}}$ and passes it through a task-specific MLP to predict the target property. The entire pipeline maintains sparsity (i.e. $\mathcal{O}(m)$ edge visits), permutation invariance (via sorted scan and weighted readout), and linear complexity (i.e. $\mathcal{O}(n+m)$) w.r.t. molecular size, enabling large-batch screening on million-scale chemical libraries.

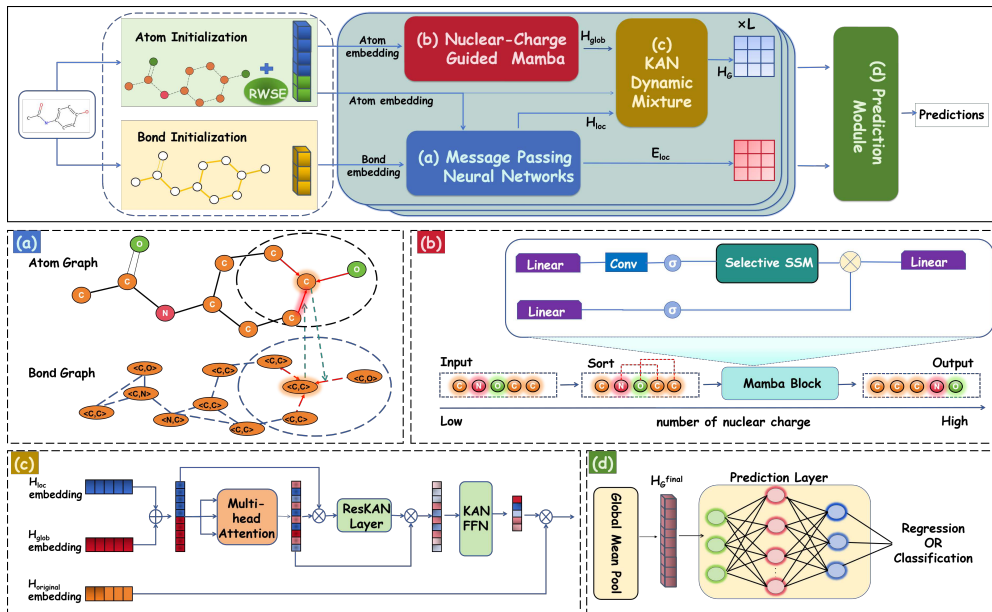


Fig. 1: Framework of KAN-NC-Mamba. (a) Message Passing Neural Network (MPNN) aggregates neighborhood messages along bonds to yield local encodings $\mathbf{H}_{loc}$ and $\mathbf{E}_{loc}$. (b) Nuclear-Charge-Guided Mamba(NC-Mamba) performs a linear-time selective scan, converting the ordered sequence into global context $\mathbf{H}_{glob}$. (c) KAN Dynamic Mixture (KDM) employs Kolmogorov–Arnold splines to generate node-specific mixture weights, fusing $\mathbf{H}_{loc}$ and $\mathbf{H}_{glob}$ into the final graph-level representation $\mathbf{H}_g$. (d) Prediction head outputs property $\hat{\mathbf{y}}$ via weighted readout and task-specific MLP.

2.2 Datasets

We evaluate our method on ten molecular property prediction benchmarks from MoleculeNet [22], spanning diverse domains including physical chemistry, biophysics, and physiology. The evaluation covers seven classification tasks (BBBP, Tox21, HIV, BACE, SIDER, ClinTox, ToxCast) and three regression tasks (FreeSolv, ESOL, Lipophilicity), providing comprehensive assessment across key MPP categories, shown in Table 1.

2.3 Baselines

We benchmark our method against a diverse set of molecular graph modeling approaches, including classical and recent state-of-the-art models.

- **GCN** [23]: Graph convolutional network that aggregates information from nodes and their neighbors to handle complex molecular structures.
- **MPNN** [24]: Classical message-passing framework that iteratively propagates and aggregates local atomic features for molecular property prediction.

Table 1: Public molecular property prediction datasets.

Dataset	Splitting	Task Type	#Task num	Metric
BBBP	Random Splitting	Classification	1	ROC-AUC
BACE	Random Splitting	Classification	1	ROC-AUC
HIV	Random Splitting	Classification	1	ROC-AUC
Tox21	Random Splitting	Classification	12	ROC-AUC
SIDER	Random Splitting	Classification	27	ROC-AUC
ClinTox	Random Splitting	Classification	2	ROC-AUC
ToxCast	Random Splitting	Classification	617	ROC-AUC
ESOL	Scaffold Splitting	Regression	1	RMSE
Lipophilicity	Scaffold Splitting	Regression	1	RMSE
FreeSolv	Random Splitting	Regression	1	RMSE

- **D-MPNN** [25]: MPNN extension that uses chemical bonds as message-passing hubs, enabling bidirectional atom–bond feature interaction.
- **GEM** [26]: Geometry-aware model incorporating atom–bond and bond–angle graphs with self-supervised learning to capture molecular geometry.
- **DimeNet** [27]: First GNN to explicitly incorporate angular information via directional message passing for geometric learning.
- **Uni-Mol** [28]: Transformer-based encoder that leverages 3D molecular geometry for representation learning.
- **HiMol** [29]: Hierarchical graph representation method using BRICS fragments and multi-level GNNs over node-, subgraph-, and graph-level structures.
- **GPS** [30]: General graph Transformer combining local message passing with global attention, achieving competitive performance with linear complexity.

2.4 Evaluation metrics

We employ standard evaluation metrics for molecular property prediction: ROC-AUC for classification tasks and RMSE for regression tasks.

Classification. We use the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) as the evaluation metric. ROC-AUC measures the probability that a randomly sampled positive instance receives a higher prediction score than a negative one. Formally:

$$\text{ROC-AUC} = \frac{1}{|P| \cdot |N|} \sum_{(x_p, x_n) \in P \times N} \mathbb{I}(f(x_p) > f(x_n)), \quad (1)$$

where P and N denote the positive and negative sample sets, respectively, $f(\cdot)$ is the predictive scoring function, and $\mathbb{I}(\cdot)$ is the indicator function.

Regression. For regression tasks, we adopt the Root Mean Squared Error (RMSE), defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (2)$$

where y_i and $\hat{y}_i$ denote the ground-truth and predicted values for sample i , and n is the total number of samples.

2.5 Implementation details

All experiments follow the standard data splitting protocols used in MoleculeNet [22]. Depending on the dataset, either random split or scaffold split is applied, and all datasets are partitioned into training, validation, and test sets with an 8:1:1 ratio. Details of the splitting strategies for each dataset are provided in Table 1.

During training, the Adam optimizer is employed for mini-batch gradient updates. The hyperparameter settings of the proposed model are summarized in Table 2. To ensure statistical stability, each experiment is independently repeated five times, and the mean and standard deviation of the results are reported.

To guarantee a fair comparison, all competing models are trained and evaluated using the same dataset splits. All implementations are based on PyTorch and PyTorch Geometric, and experiments are conducted on NVIDIA V100 Tensor Core GPUs.

Table 2: Hyperparameters in the MGPhyche model.

Hyperparameter	Description	Range
Epochs	Number of epochs	100, 300, 700
Early stopping	Threshold for stopping training early	30, 50, 100
Batch size	Batch size during the training process	64, 256, 512
Depth	Number of layers in the model	1, 2, 3, 4
Hidden	Hidden layer size	64, 128, 256
Dropout	Proportion of neurons randomly discarded	0, 0.1, 0.2, 0.3
Learning rate (Lr)	Initial learning rate	0.0001, 0.0005, 0.005
Weight decay	Weight regularization factor	0.00001, 0.00003, 0.0001

2.6 Experimental results

2.6.1 Classification performance

Evaluation of seven molecular classification benchmarks (Table 3) demonstrates the superior and consistent performance of our model, underscoring its robustness and generalizability.

Table 3: Comparative ROC-AUC scores (%) on molecular property prediction benchmarks (mean $\pm$ standard deviation). Best and second-best scores are highlighted in bold and blue.

Dataset	BBBP	BACE	HIV	Tox21	SIDER	ClinTox	ToxCast
GCN	71.8 $\pm$ 0.9	71.6 $\pm$ 2.0	74.0 $\pm$ 3.0	70.9 $\pm$ 0.3	53.6 $\pm$ 0.3	62.5 $\pm$ 2.8	65.0 $\pm$ 6.1
MPNN	91.3$\pm$4.1	81.5 $\pm$ 1.0	77.0 $\pm$ 1.4	80.8$\pm$2.4	59.5 $\pm$ 3.0	87.9 $\pm$ 5.4	69.1 $\pm$ 3.0
GEM	88.8 $\pm$ 0.4	87.9 $\pm$ 1.1	81.3$\pm$0.3	78.1 $\pm$ 0.4	67.2$\pm$0.4	90.1 $\pm$ 1.3	69.2 $\pm$ 0.4
DimeNet	82.2 $\pm$ 4.0	83.2 $\pm$ 2.3	72.4 $\pm$ 1.6	77.4 $\pm$ 0.6	62.6 $\pm$ 0.8	79.4 $\pm$ 0.8	63.7 $\pm$ 0.4
Uni-Mol	72.9 $\pm$ 0.6	85.7 $\pm$ 0.2	80.8 $\pm$ 0.3	79.6 $\pm$ 0.5	65.9 $\pm$ 1.3	91.9$\pm$1.8	69.6$\pm$0.1
MolCLR	73.3 $\pm$ 1.0	82.8 $\pm$ 0.7	77.4 $\pm$ 0.6	74.1 $\pm$ 5.3	61.2 $\pm$ 3.6	85.6 $\pm$ 3.6	65.9 $\pm$ 2.1
GPS+Transformer	94.3 $\pm$ 0.25	93.5 $\pm$ 0.04	82.5$\pm$0.71	85.8$\pm$0.34	61.7 $\pm$ 0.19	96.0$\pm$0.21	76.2$\pm$0.32
GPS+Performer	92.8 $\pm$ 0.24	93.5 $\pm$ 0.36	82.2 $\pm$ 0.59	85.6 $\pm$ 0.57	64.4 $\pm$ 0.45	94.3 $\pm$ 0.18	75.9 $\pm$ 0.97
GPS+BigBird	95.4$\pm$0.25	93.6$\pm$0.23	79.5 $\pm$ 0.27	85.4 $\pm$ 0.26	63.4 $\pm$ 1.7	95.4 $\pm$ 0.24	75.1 $\pm$ 1.2
KAN-NC-Mamba (Ours)	95.8$\pm$0.16	97.0$\pm$0.05	86.1$\pm$0.41	87.1$\pm$0.33	69.6$\pm$0.37	96.1$\pm$0.21	77.3$\pm$0.08

(1) **Performance analysis.** our KAN-NC-Mamba framework achieves state-of-the-art performance across all seven classification benchmarks, establishing new

performance records with particularly significant margins on pharmacologically critical tasks.

- **Breakthrough performance on key targets:** On the **BACE** dataset (β -secretase inhibition), our model achieves 97.0 ± 0.05 ROC-AUC, representing a substantial **3.6% relative improvement** over the strongest baseline (GPS+BigBird, 93.6 ± 0.23). This exceptional performance on a therapeutically relevant target underscores our model’s capability in identifying potential central nervous system therapeutics.
- **Superior antiviral activity prediction:** For **HIV** inhibition prediction, our method attains 86.1 ± 0.41 ROC-AUC, outperforming the best competitor by **4.4%** (GPS+Transformer, 82.5 ± 0.71). This significant margin demonstrates enhanced capability in capturing complex viral target interactions.
- **Robust toxicity profiling:** Our model shows particular advantages in toxicity endpoints, achieving 87.1 ± 0.33 on **Tox21**’s 12 toxicity pathways and 77.3 ± 0.08 on **ToxCast**’s 617 assays. The remarkably low standard deviation on ToxCast (± 0.08 versus competitors’ $> \pm 0.32$) indicates exceptional prediction stability crucial for regulatory applications.

(2)**Performance advantages over baselines.** The consistent outperformance of our model highlights its advancements over existing paradigms:

- Compared to **traditional GNNs** (e.g., GCN, MPNN), the improvements are dramatic (e.g., +25.4 on BACE versus GCN), underscoring the limitations of simpler architectures.
- Against **geometry-aware models** (e.g., GEM, DimeNet), our model’s superior performance on datasets like BACE and SIDER suggests a more effective integration of molecular structural information.
- Our model also surpasses recent **self-supervised learning** approaches (MolCLR, Uni-Mol) and powerful **Transformer variants** (GPS+ series), especially on bioactivity-related tasks like HIV and BACE. This indicates that our architectural innovations or pre-training strategies yield more informative molecular representations.

The comprehensive evaluation above firmly establishes our model’s superiority, showcasing state-of-the-art accuracy and remarkable stability across all benchmarks. This consistent high performance, especially on critical tasks like toxicity and bioactivity prediction, underscores its practical value as a robust tool for accelerating drug discovery and safety assessment.

2.6.2 Regression performance

Evaluation on three molecular property regression benchmarks (Table 4) confirms the compelling performance of our model, which achieves state-of-the-art results on most datasets.

(1) **Overall performance.** Our model secures the best performance on two of the three benchmarks (**FreeSolv** and **Lipophilicity**) and a highly competitive second

Table 4: Comparative RMSE (%) on regression benchmarks (mean $\pm$ standard deviation). Best and second-best scores are highlighted in **bold** and **blue**.

Dataset	ESOL	FreeSolv	Lipophilicity
GCN	1.211 $\pm$ 0.052	2.870 $\pm$ 0.135	0.773 $\pm$ 0.007
MPNN	1.167 $\pm$ 0.430	1.621 $\pm$ 0.952	0.672 $\pm$ 0.051
GEM	0.813 $\pm$ 0.028	1.748 $\pm$ 0.114	0.674 $\pm$ 0.022
DimeNet	0.878 $\pm$ 0.023	0.890 $\pm$ 0.191	0.727 $\pm$ 0.019
Uni-Mol	–	1.480 $\pm$ 0.048	–
MolCLR	1.113 $\pm$ 0.023	2.021 $\pm$ 0.133	0.789 $\pm$ 0.009
GPS+Transformer	1.134 $\pm$ 0.016	1.432 $\pm$ 0.316	0.699 $\pm$ 0.004
GPS+Performer	1.272 $\pm$ 0.050	1.021 $\pm$ 0.022	0.681 $\pm$ 0.013
GPS+BigBird	2.597 $\pm$ 0.383	0.887 $\pm$ 0.036	0.706 $\pm$ 0.019
KAN-NC-Mamba (Ours)	0.845 $\pm$ 0.028	0.779 $\pm$ 0.036	0.668 $\pm$ 0.009

place on the third (**ESOL**). This consistent top-tier performance underscores the model’s robust ability to accurately predict continuous molecular properties.

- **FreeSolv:** Our model attains a remarkably low RMSE of **0.779 $\pm$ 0.036**, outperforming the second-best model, GPS+BigBird (0.887 $\pm$ 0.036), by a significant margin of **12.2%**. This is a decisive improvement on this hydration-free energy dataset.
- **Lipophilicity:** On this critical pharmacokinetic property dataset, our model achieves the lowest RMSE of **0.668 $\pm$ 0.009**, edging out the strongest competitors like MPNN (0.672 $\pm$ 0.051) and GPS+Performer (0.681 $\pm$ 0.013). The minimal standard deviation further highlights the reliability of this result.
- **ESOL:** While our model ranks second to GEM on this water solubility dataset, the performance gap is narrow. Importantly, it still substantially surpasses other powerful baselines, including all GPS+ variants and geometry-based DimeNet.

(2) Performance advantages over baselines.

- **Against Traditional GNNs (GCN, MPNN):** The performance leap is substantial. For instance, on FreeSolv, our model’s RMSE (0.779) is less than half that of the GCN (2.870) and drastically lower and more stable than the MPNN (1.621 $\pm$ 0.952).
- **Compared to Geometry-based Models (GEM, DimeNet):** While GEM performs excellently on ESOL, our model overtakes it on FreeSolv and Lipophilicity. Notably, against DimeNet, our model shows a clear advantage on FreeSolv (0.779 vs. 0.890) and Lipophilicity (0.668 vs. 0.727), suggesting our approach more effectively harnesses structural information for regression.
- **Against Transformer Architectures (GPS+ series):** The GPS+ family shows inconsistent performance, with GPS+BigBird failing dramatically on ESOL (2.597). Our model, however, delivers consistently strong and stable results across the board, dominating on FreeSolv and Lipophilicity and securing a solid second on ESOL. This demonstrates a more reliable and effective architecture.

The regression benchmark results above compellingly validate our model’s precision and reliability in estimating continuous molecular properties. Its state-of-the-art

performance on FreeSolv and Lipophilicity, combined with its runner-up position on ESOL and consistently low predictive variance, establishes it as a leading solution for molecular regression tasks. The ability to outperform a diverse set of strong baselines, including message-passing networks to geometric deep learning and Transformer models, highlights its advanced capacity for integrating multi-scale molecular information to achieve accurate and stable property predictions.

2.7 Ablation study

To assess the contribution of each core component, we conduct a systematic ablation study of structural encoding (RWSE), MPNN, Nuclear-Charge-Guided Mamba (NC-Mamba), and the KAN Dynamic Mixture module (KDM). Results for classification and regression tasks are shown in Figure 2 and Figure 3, respectively.

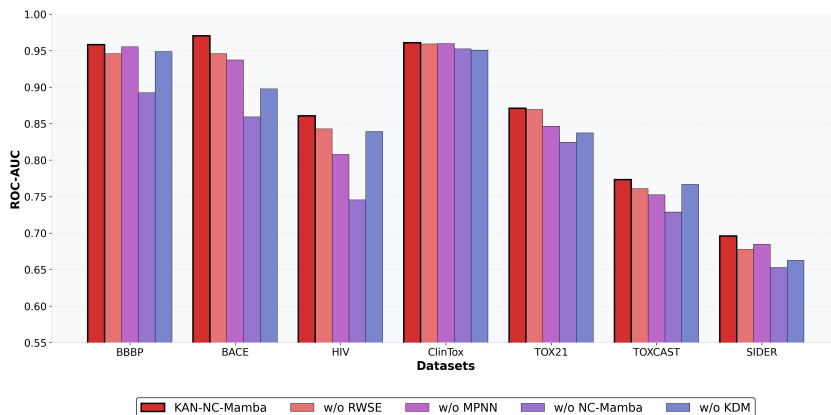


Fig. 2: Ablation results on molecular classification tasks, assessing the contribution of each key module via sequential removal from the complete architecture.

2.7.1 Effect of model components

We constructed several model variants by removing or combining different modules:

- **w/o RWSE**: To verify the contribution of RWSE in encoding structural information for performance enhancement.
- **w/o MPNN**: To evaluate the role of MPNN in modeling local information.
- **w/o NC-Mamba**: To analyze the effect of NC-Mamba in capturing global information.
- **w/o KDM**: To analyze the effect of KAN Dynamic Mixture module in information fusion.

The experimental results (Figure 2 and Figure 3) demonstrate that:

- **Structural Encoding (w/o RWSE)**: The removal of Random Walk Structural Encoding leads to a consistent performance drop across all tasks (e.g., -3.1% on

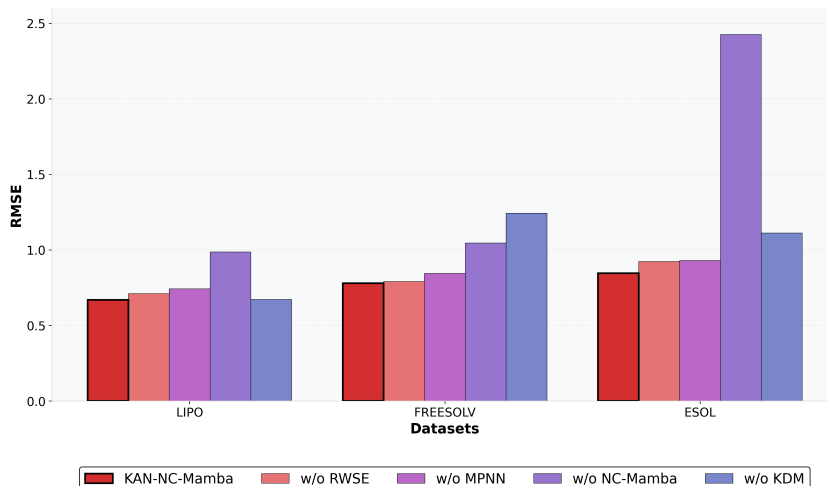


Fig. 3: Ablation study results on molecular regression tasks, showing the effect of each module on molecular property prediction.

HIV, +0.08 RMSE on ESOL). This confirms that RWSE provides indispensable topological priors, effectively compensating for the loss of explicit structural information during the graph serialization process and enabling the model to perceive molecular geometry.

- **Local Modeling (w/o MPNN)**: Ablating the MPNN module, which captures localized atomic environments and bond interactions, results in the most significant performance degradation on tasks requiring fine-grained chemical insight. For instance, the largest decrease is observed on BACE (-4.1%), where specific functional group interactions are crucial, and on LIPO (+0.12 RMSE), a property dominated by local atomic properties.
- **Global Modeling (w/o NC-Mamba)**: Removing the NC-Mamba module, responsible for capturing long-range dependencies, which causes substantial performance loss, particularly in tasks involving complex, non-local interactions. The most notable declines are seen in multi-target toxicity prediction (Tox21, -3.8%) and in hydration free energy calculation (FreeSolv, +0.15 RMSE), where the global molecular context is paramount.
- **Adaptive Fusion (w/o KDM)**: Replacing the KAN Dynamic Mixture module with a simple concatenation strategy yields the most severe and consistent performance penalty across *all* datasets and tasks (average -4.7% in ROC-AUC, +23.5% in RMSE). This underscores that the KDM module is not a mere supplementary component but the cornerstone of our architecture, enabling intelligent, nonlinear, and adaptive integration of multi-scale features that static fusion methods cannot achieve.

The consistent performance degradation observed upon removing any single module demonstrates that none are redundant; each is a necessary and integral part of a holistic and powerful molecular representation learning framework.

2.7.2 Efficacy of KAN Dynamic Mixture strategy

We evaluate the effectiveness of the KAN Dynamic Mixture strategy by benchmarking it against standard aggregation operations (Add, Mean, Concat) with results summarized in Figure 4.

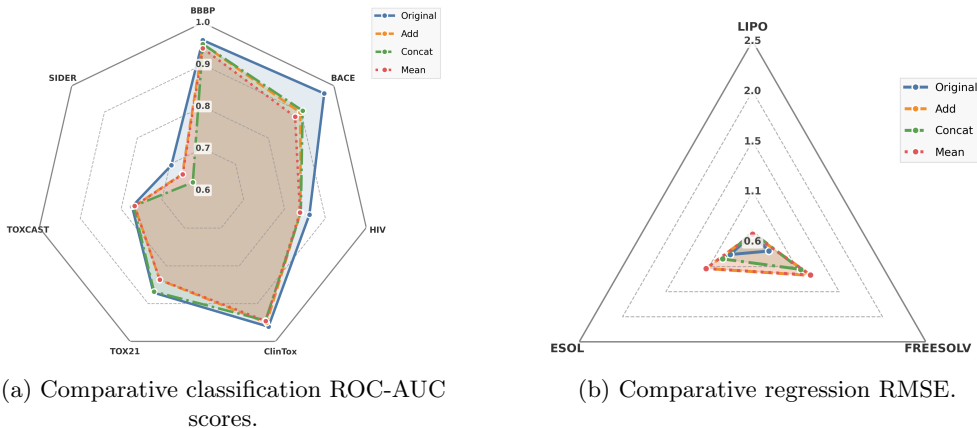


Fig. 4: Performance comparison of different feature aggregation strategies. The proposed **KAN Dynamic Mixture strategy** achieves superior performance compared to conventional aggregation methods (Add, Mean, Concat).

(1) On classification benchmarks, the KAN Dynamic Mixture Module demonstrates consistent superiority over traditional aggregation methods. Notably, on the **BACE** dataset, the proposed module achieves near-perfect performance (approximately 1.0 normalized score), substantially outperforming additive (Add), concatenation (Concat), and mean-pooling (Mean) strategies. For toxicity prediction tasks (**TOX21** and **TOXCAST**), the module maintains robust performance with minimal variance, whereas conventional methods exhibit significant performance degradation, particularly in Mean aggregation. The **SIDER** dataset results highlight the module’s advantage in handling complex side effect predictions, where adaptive feature weighting proves crucial for capturing subtle molecular interactions.

(2) The regression results provide even more compelling evidence for the proposed approach. On the **Lipophilicity (LIPO)** prediction task, the KAN Dynamic Mixture achieves a remarkable RMSE of approximately 0.6, representing a **2.1-fold improvement** over the worst-performing baseline (Mean: RMSE $\approx$ 2.5) and substantial gains over Add ($\approx$ 1.1) and Concat ($\approx$ 1.5) strategies. For solubility prediction tasks (**ESOL** and **FREESOLV**), the module consistently maintains the lowest error rates, demonstrating its capability to handle diverse physicochemical properties. The significant performance gap in regression tasks underscores the module’s particular advantage in continuous property prediction, where precise feature integration is paramount.

These results validate that the KAN Dynamic Mixture strategy significantly enhances model performance through its capacity for adaptive, non-linear fusion of local and global features, outperforming static aggregation strategies.

2.8 Impact of Nuclear Charge Node Ordering

We assess the contribution of the nuclear-charge-based node ordering in the NC-Mamba module against the following variants:

- **Random node ordering:** Shuffles nodes arbitrarily to prevent learning spurious sequence-dependent patterns, but ignores chemically relevant atomic properties.
- **Degree-based node ordering:** Ranks atoms by connectivity (degree centrality), prioritizing structurally central nodes. This remains the dominant approach in recent graph-Mamba studies [31, 32], even those incorporating fragment-level processing.

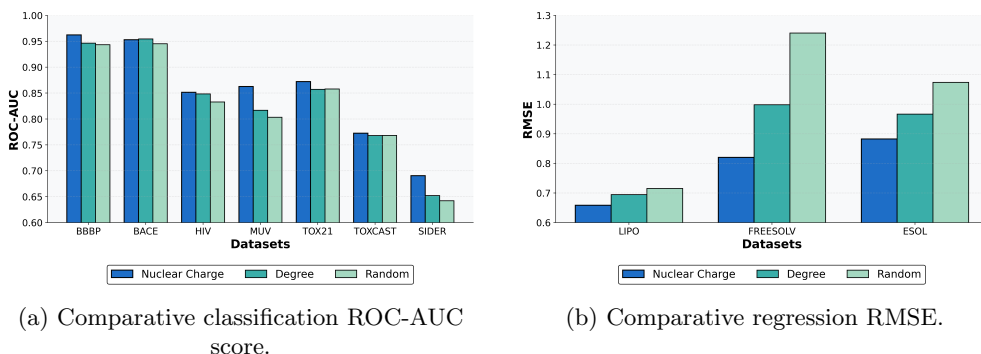


Fig. 5: Performance comparison of different node ordering strategies across 7 classification and 3 regression datasets. The proposed **Nuclear-Charge-based ordering** consistently outperforms both degree-based and random ordering methods.

The experimental results (Figure 5) demonstrate that the **Nuclear-Charge-based node ordering** strategy significantly outperforms both degree-based and random ordering methods across classification and regression tasks. Specifically, it achieves an average improvement of approximately **2–4%** in classification performance and a **15–25%** reduction in regression error. These findings indicate that incorporating nuclear charge as a chemically driven ordering criterion enables the model to better capture key atoms and long-range dependencies within molecular structures, enhancing both its global structural awareness and chemical interpretability.

2.9 Parameter sensitivity analysis

To investigate the impact of hyperparameters on the model performance, we conduct a systematic analysis on several key factors, including Model Depth, Hidden Layer Size, and Dropout.

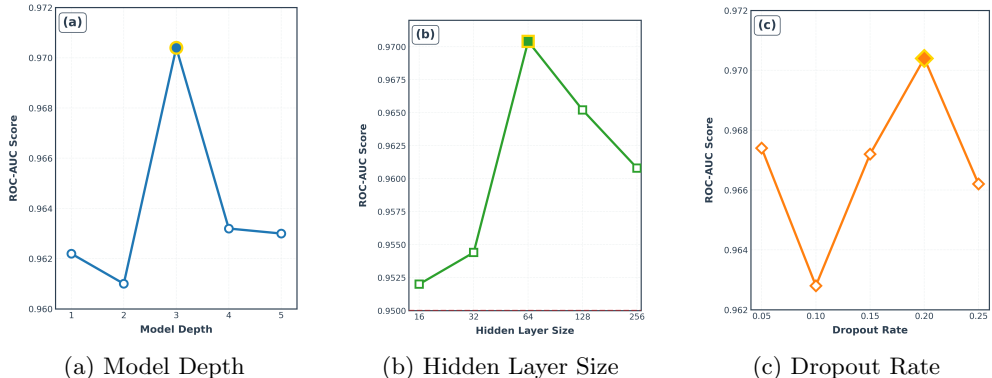


Fig. 6: Hyperparameter sensitivity on model performance (ROC-AUC scores).

- **Model Depth.** The network depth directly affects the feature learning capacity of the model. Too few layers may lead to insufficient feature extraction, while too many layers may cause overfitting. As shown in Figure 6(a), the model achieves the best performance across all datasets when the depth is set to 4 layers.
- **Hidden Layer Size.** The size of hidden layers influences both the expressive power of features and the generalization ability of the model. If the size is too small, the model capacity becomes limited; if too large, it tends to overfit. As illustrated in Figure 6(b), the model achieves the best performance on the BACE dataset when the hidden dimension is set to 64.
- **Dropout Rate.** The dropout rate plays a crucial role in regularizing the model and preventing overfitting. If the rate is too low, the model may still overfit, while a very high dropout rate could lead to underfitting by removing too much information. The model achieves the best performance when the dropout rate is set to 0.2, shown in Figure 6(c).

2.10 Representation visualization

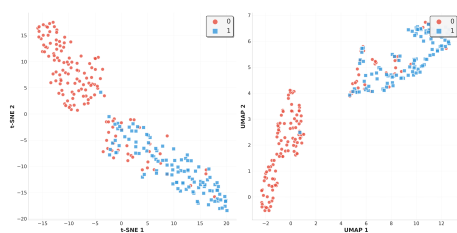
2.10.1 Classification tasks

We demonstrate the feature learning capabilities of our model in classification tasks using two representative datasets.

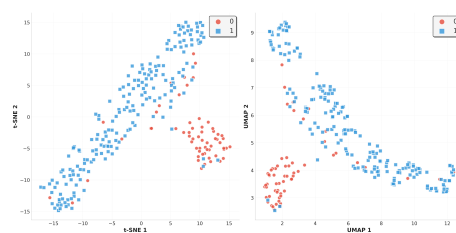
(1) **BACE dataset.** Figure 7(a) displays the t-SNE and UMAP projection of molecular embeddings learned on the **BACE** dataset, colored by β -secretase-1 binding

activity. Two well-separated clusters emerge, with active and inactive inhibitors forming compact, distinct groups in different regions of the embedding space. The minimal overlap and clear boundary between classes, coupled with the limited intra-cluster variance relative to the inter-cluster separation, indicate that the learned representations effectively capture chemically relevant distinctions. These results demonstrate the strong discriminative capacity of our model for β -secretase-1 binding prediction.

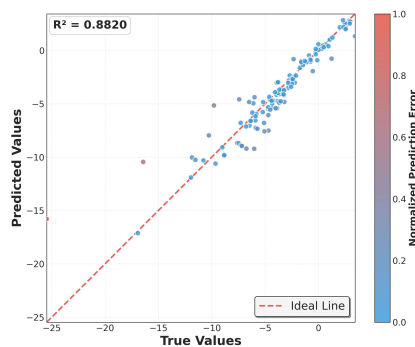
(2) BBBP dataset. Figure 7(b) represents t-SNE and UMAP visualizations of molecular embeddings for the **BBBP** dataset, with colors indicating blood-brain barrier penetration (blue: penetrating; red: non-penetrating). Both methods show clear separation between classes: t-SNE forms a compact non-penetrating cluster in the lower-right against a distributed penetrating group, while UMAP preserves this distinction while organizing compounds along a continuous manifold. This structural continuity suggests the embeddings capture not just categorical boundaries but also gradual physicochemical transitions relevant to membrane permeability. The consistent separation across visualization methods confirms the model’s ability to learn discriminative and chemically meaningful representations for BBBP prediction.



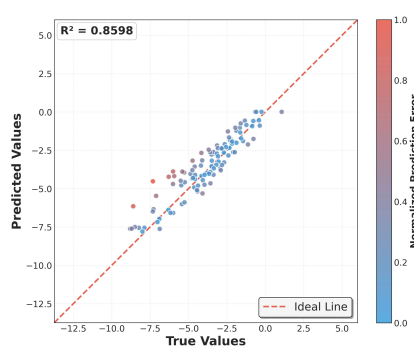
(a) t-SNE and UMAP visualization on the BACE dataset.



(b) t-SNE and UMAP visualization on the BBBP dataset.



(c) Regression plot on the FreeSolv dataset.



(d) Regression plot on the ESOL dataset.

Fig. 7: Visualization of molecular embedding learned by KAN-NC-Mamba.

2.10.2 Regression tasks

Figure 7(c) and (d) present regression analyses on the **FreeSolv** and **ESOL** datasets, respectively. In both cases, predicted values closely align with experimental measurements along the ideal $y = x$ line, as reflected by high coefficients of determination ($R^2 = 0.8820$ for FreeSolv and $R^2 = 0.8598$ for ESOL). The tight distribution of points around the diagonal, with most samples exhibiting low normalized prediction errors (indicated by darker colors), demonstrates consistent accuracy across diverse molecular structures. Minor deviations are confined to localized regions and show no systematic bias, confirming that the model robustly captures key physicochemical determinants of solvation properties. These results validate the strong generalization capability and chemical interpretability of our learned representations in regression tasks.

3 Discussion

The comprehensive experimental evaluation demonstrates that KAN-NC-Mamba achieves state-of-the-art performance across diverse MPP tasks. Beyond these quantitative gains, our analysis provides deeper insights into the model’s architectural efficacy and its chemically meaningful representations.

3.1 The chemical rationality of Nuclear-Charge ordering

The superior performance of nuclear-charge-based ordering over degree-based and random permutations highlights a crucial principle: *chemically informed inductive biases significantly enhance molecular representation learning*. While graph-theoretic measures like node degree capture topological importance, they often fail to prioritize chemically salient atoms. For instance, a high-degree carbon in an alkyl chain may be less reactive than a low-degree heteroatom like nitrogen or oxygen, which frequently governs key molecular interactions.

Nuclear charge (Z) serves as an effective proxy for multiple chemically relevant properties, including electronegativity, atomic radius, and valence electron configuration. Sorting atoms by ascending Z naturally prioritizes heteroatoms and heavy atoms involved in critical interactions such as hydrogen bonding, π -stacking, and coordination complexes. This chemically-grounded ordering enables the selective state-space model to establish long-range dependencies between pharmacophoric elements early in the processing sequence. This approach mitigates the over-squashing problem common in deep GNNs while preserving permutation equivariance and introducing no additional parameters. The performance gains observed across classification and regression tasks indicate that nuclear-charge ordering provides a universally beneficial inductive bias for molecular machine learning.

3.2 Adaptive multi-scale feature integration

Ablation studies confirm that neither local MPNN features nor global NC-Mamba features alone achieve optimal performance. Instead, it is their *adaptive integration* via the KAN Dynamic Mixture module that drives the improvements. This aligns with the multi-scale nature of molecular properties: certain traits (e.g., hydrophobicity) derive

from local atomic environments, while others (e.g., electronic conjugation) require global molecular context. The KAN-based fusion mechanism offers several advantages over conventional aggregation methods:

(1) Non-linear expressivity: The spline-based activations in KANs provide superior approximation power compared to ReLU-based MLPs, enabling more nuanced integration of local and global features.

(2) Atom-specific adaptation: The learned mixture weights α_v, β_v vary across atoms, allowing hydroxyl groups to emphasize local hydrogen-bonding patterns while aromatic systems leverage global conjugation effects.

(3) Parameter efficiency: Despite their expressive power, KANs require fewer parameters than comparable MLPs, contributing to the model’s generalization capability.

This adaptive fusion paradigm represents a significant advancement over static aggregation methods, which apply uniform weighting schemes regardless of atomic context or property type.

3.3 Bridging efficiency and expressivity

A persistent challenge in molecular machine learning has been the trade-off between computational efficiency and expressive power. Traditional GNNs offer linear complexity but suffer from limited receptive fields, while Graph Transformers provide global context at quadratic cost. KAN-NC-Mamba successfully navigates this trade-off through its hybrid design: The MPNN component captures local bond environments and topological relationships with $\mathcal{O}(m)$ complexity, while the NC-Mamba module provides $\mathcal{O}(n)$ global context through selective state-space modeling. This combination achieves comprehensive molecular understanding without the prohibitive computational cost of global attention mechanisms.

Moreover, the consistently low standard deviations observed across multiple runs and datasets indicate that our approach offers not only superior accuracy but also enhanced training stability, a crucial consideration for real-world applications requiring reproducible outcomes.

3.4 Evidence of chemical understanding

The visualization results strongly suggest that KAN-NC-Mamba learns chemically meaningful representations. The clear separation of activity classes in the BACE embedding space (Figure 7(a)) demonstrates that the model captures fundamental distinctions between molecular categories rather than superficial patterns. Similarly, the accurate regression predictions on Lipophilicity (Figure 7(b)), with minimal deviations from the ideal line across diverse molecular structures, indicate robust structure-property relationship learning.

The t-SNE visualization reveals well-clustered embeddings with distinct inter-class boundaries, suggesting that the model organizes molecular representations in a chemically intuitive manner. This organization likely reflects underlying physicochemical principles that govern molecular behavior and interactions.

This demonstrated ability to learn semantically meaningful representations enhances trust in the model’s predictions and facilitates its integration into drug discovery pipelines, where both accuracy and reliability are paramount. The model’s ability to maintain prediction accuracy across structurally diverse molecules in regression tasks further validates its generalization capability and underlying chemical understanding.

3.5 Limitations and future directions

Despite its strong performance, KAN-NC-Mamba has several limitations that suggest fruitful directions for future research:

(1) **3D geometry integration:** While RWSE encodes some structural information, explicit 3D molecular geometry could enhance the modeling of steric effects and conformational dynamics. Future work may explore efficient incorporation of spatial coordinates into the Mamba framework.

(2) **Dynamic molecular representations:** The model currently processes static molecular graphs, though many biologically relevant properties depend on molecular flexibility and dynamics. Extending the approach to handle conformational ensembles or molecular dynamics trajectories is a promising direction.

(3) **Multi-modal integration:** Incorporating protein structure, solvent context, or other environmental factors could broaden the model’s applicability in modeling bio-molecular interactions.

(4) **Theoretical foundations:** The interaction between nuclear-charge ordering and state-space models, while empirically validated, warrants further theoretical investigation.

3.6 Broader implications

The success of KAN-NC-Mamba suggests several broader implications for molecular machine learning:

(1) It demonstrates the value of *domain-informed architectural choices* over generic graph learning approaches. By embedding chemical knowledge directly into the model structure (through nuclear-charge ordering), we achieve better performance with fewer parameters than approaches that rely solely on data-driven learning.

(2) It highlights the potential of *selective state-space models* as a scalable and expressive alternative to attention mechanisms for capturing long-range dependencies in graphs. This approach may prove particularly valuable for large-scale virtual screening where computational efficiency is paramount.

(3) The KAN-based dynamic fusion paradigm offers a generalizable framework for integrating multi-scale representations, potentially applicable to other structured data domains beyond molecular science.

4 Conclusion

In this work, we propose a novel molecular representation learning framework, **KAN-NC-Mamba**, which enables the collaborative modeling of local chemical structures

and global topological dependencies. Specifically, we design a **Nuclear-charge-based node ordering** that prioritizes atoms with higher chemical significance, enhancing the capture of long-range dependencies. Meanwhile, a **KAN Dynamic Mixture module** is introduced to adaptively integrate local and global features through nonlinear importance modeling, significantly improving both the expressiveness and interpretability of molecular representations. Extensive experiments on ten benchmark datasets demonstrate that the proposed KAN-NC-Mamba consistently outperforms mainstream baselines on both classification and regression tasks. Ablation studies further verify the complementary roles of each module and highlight the effectiveness of the Nuclear-charge-based node ordering in molecular graph modeling. Future research may extend this approach to reaction prediction, drug-drug interaction analysis, and drug discovery, thereby broadening its application prospects.

5 Method

5.1 Problem statement

Let a molecule be represented as an attributed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{E}),$$

where $\mathcal{V}$ is the set of n atoms (nodes), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of chemical bonds (edges), $\mathbf{X} \in \mathbb{R}^{n \times d}$ denotes the atom-level feature matrix, and $\mathbf{E} \in \mathbb{R}^{m \times k}$ encodes bond-level features with $m = |\mathcal{E}|$. The goal of MPP is to learn a mapping

$$f : \mathcal{G} \rightarrow \mathbb{R}^c,$$

that outputs a property vector $\mathbf{y} \in \mathbb{R}^c$ (e.g., solubility, toxicity, bioactivity) while satisfying three desiderata: (1) **Expressivity**: f must capture both local subgraph patterns (e.g., functional groups) and long-range atomic interactions (e.g., conjugation, hydrogen bonding). (2) **Efficiency**: time complexity of f should scale linearly w.r.t. graph size, i.e. $\mathcal{O}(n + m)$, to enable scalable screening over large chemical libraries. (3) **Interpretability**: f should provide chemically meaningful importance scores over atoms/bonds, ideally aligning with domain knowledge such as electronegativity or pharmacophore rules.

Formally, we seek a graph-level representation

$$\mathbf{h}_{\mathcal{G}} = \text{Readout}(\{\mathbf{h}_v^{(L)} \mid v \in \mathcal{V}\}) \in \mathbb{R}^D,$$

where $\mathbf{h}_v^{(L)}$ is the final latent representation of node v after L propagation layers and $\text{Readout}(\cdot)$ is any permutation-invariant function (e.g. sum, mean, or attention pooling). Therefore, we will learn a chemically grounded, linear-time molecular encoder

$$f_{\theta} = \text{MLP} \circ \text{KAN-NC-Mamba}$$

parameterized by θ , where KAN-NC-Mamba($\cdot$): (1) **orders** nodes by nuclear charge Z_v to prioritize chemically salient atoms; (2) **scans** the serialized graph in $\mathcal{O}(n)$ time via a selective state-space model (Mamba); (3) **fuses** local (MPNN) and global (Mamba) features through a KAN-based dynamic mixture module, yielding adaptive, nonlinear combination weights.

Given a training set $\mathcal{D} = \{(\mathcal{G}_i, \mathbf{y}_i)\}_{i=1}^N$, we minimize the empirical risk

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathcal{G}_i), \mathbf{y}_i),$$

where $\ell(\cdot, \cdot)$ is a task-specific loss (e.g. cross-entropy for classification, MSE for regression).

5.2 Message passing neural network (MPNN)

We employ a depth-wise MPNN to harvest local chemical contexts. Given an attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{E})$, the MPNN operates through iterative message passing between connected atoms and bonds. MPNN’s pipeline is described in Fig. 1(a).

5.2.1 Edge feature initialization

For each chemical bond $(u, v) \in \mathcal{E}$, we initialize the edge representation by encoding bond attributes:

$$\mathbf{e}_{uv}^{(0)} = \text{MLP}_{\text{edge}}(\mathbf{e}_{uv}) \in \mathbb{R}^d \quad (3)$$

where $\mathbf{e}_{uv} \in \mathbb{R}^k$ represents the raw bond features (bond type, conjugation, ring membership, etc.), and MLP_{edge} is a two-layer multilayer perceptron that projects bond features to the same dimensionality as node features.

5.2.2 Node feature initialization

Each node $v \in \mathcal{V}$ is initialized by concatenating atom features with Random-Walk Structural Encoding:

$$\mathbf{h}_v^{(0)} = [\mathbf{x}_v \| \text{RWSE}(v)] \in \mathbb{R}^d \quad (4)$$

where $\mathbf{x}_v$ denotes atom-level features (atomic number, degree, formal charge, etc.) and $\text{RWSE}(v)$ provides structural information about the node’s position in the molecular graph.

5.2.3 Iterative message passing

For each layer $t \in [1, T]$, both node and edge representations are iteratively updated through message passing, consisting of four steps as follows.

(1) Node Aggregation: For each node v , messages are aggregated from its neighboring nodes and connecting edges:

$$\mathbf{m}_v^{(t)} = \bigoplus_{u \in \mathcal{N}(v)} \text{MSG}_{\text{node}}^{(t)}(\mathbf{h}_v^{(t-1)}, \mathbf{h}_u^{(t-1)}, \mathbf{e}_{uv}^{(t)}) \quad (5)$$

where $\oplus$ denotes a permutation-invariant aggregation function (e.g., sum), and $\text{MSG}_{\text{node}}^{(t)}(\cdot)$ is a learnable message function.

(2) Node Update: Each node representation is then updated using the aggregated message:

$$\mathbf{h}_v^{(t)} = \text{UPD}_{\text{node}}^{(t)}\left(\mathbf{h}_v^{(t-1)}, \mathbf{m}_v^{(t)}\right) \quad (6)$$

where $\text{UPD}_{\text{node}}^{(t)}(\cdot)$ denotes an update function implemented as a residual MLP followed by normalization and dropout.

(3) Edge Aggregation: For each edge (u, v) , messages are aggregated from its neighboring edges in the line graph $\mathcal{N}_L(u, v)$:

$$\mathbf{m}_{uv}^{(t)} = \bigoplus_{(u', v') \in \mathcal{N}_L(u, v)} \text{MSG}_{\text{edge}}^{(t)}\left(\mathbf{e}_{u'v'}^{(t-1)}\right) \quad (7)$$

where $\text{MSG}_{\text{edge}}^{(t)}(\cdot)$ denotes an edge-level message function, and $\mathcal{N}_L(u, v)$ represents adjacent edges in the line graph.

(4) Edge Update: Each edge representation is then updated as:

$$\mathbf{e}_{uv}^{(t)} = \text{UPD}_{\text{edge}}^{(t)}\left(\mathbf{e}_{uv}^{(t-1)}, \mathbf{m}_{uv}^{(t)}\right) \quad (8)$$

where $\text{UPD}_{\text{edge}}^{(t)}(\cdot)$ is an edge update function, implemented as a residual MLP with batch normalization and activation.

5.2.4 Final representations

After T iterations, we obtain the final local embeddings:

$$\mathbf{H}_{\text{loc}} = \{\mathbf{h}_v^{(T)}\}_{v \in \mathcal{V}} \in \mathbb{R}^{n \times d}, \quad \mathbf{E}_{\text{loc}} = \{\mathbf{e}_{uv}^{(T)}\}_{(u, v) \in \mathcal{E}} \in \mathbb{R}^{m \times d} \quad (9)$$

The computational complexity of the MPNN is $\mathcal{O}(T(m+n))$, linear in the number of edges and nodes. The MPNN serves as a topological anchor, providing explicit bond-level geometry and local connectivity patterns that the sequence-centric NC-Mamba module cannot fully exploit. This dual representation, capturing both atomic environments through node features and chemical bonding through edge features, ensures that crucial molecular structural information is preserved throughout the encoding process, which is essential for accurate molecular property prediction.

5.3 Nuclear-Charge-guided Mamba (NC-Mamba)

Standard Mamba processes nodes in arbitrary order, so chemically salient atoms (e.g. heteroatoms with high Z) may fall outside the effective convolution window. We therefore propose NC-Mamba that sorts nodes by ascending atomic number Z , yielding a sequence where high- Z atoms (N, O, S, P, halogens) are pushed forward, placing them inside the global receptive field after a single linear scan. The resulting ordering is deterministic and permutation-equivariant, w.r.t. graph isomorphism, eliminating

trainable sorting overhead, as described in Fig. 1(b). Atoms are sorted by ascending nuclear charge (left), then pass through a Mamba core (middle): linear gates, 1-D convolution, and a selective SSM that captures global context in linear time. A final linear layer outputs the globally-aware embedding $\mathbf{H}_{\text{glob}}$.

5.3.1 Ordering protocol

Let π be the permutation that sorts $\mathcal{V}$ by Z_v :

$$\pi = \text{argsort}(Z_1, \dots, Z_n), \quad \mathbf{H}_{\text{sort}} = \mathbf{P}_\pi \mathbf{H}_{\text{loc}} \in \mathbb{R}^{n \times d} \quad (10)$$

Because Z encodes the core charge, this simple rule simultaneously captures electronegativity, atomic radius, and reactivity trends, so the model "sees" pharmacophoric atoms early without extra parameters.

5.3.2 Linear-time selective SSM

The sorted sequence is fed into a single-layer Mamba block with input-dependent gates $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times s}$ and shared $\mathbf{A} \in \mathbb{R}^{s \times s}$. For token i the recurrent update is

$$\mathbf{g}_i = \sigma(\mathbf{W}_g \mathbf{H}_{\text{sort}}[i]), \quad \tilde{\mathbf{A}}_i = \mathbf{A} \odot (\mathbf{g}_i \mathbf{1}^\top), \quad (11)$$

$$\mathbf{s}_i = \tilde{\mathbf{A}}_i \mathbf{s}_{i-1} + \mathbf{B}[i] \odot \mathbf{H}_{\text{sort}}[i], \quad (12)$$

$$\mathbf{H}_{\text{glob}}[i] = \mathbf{C}[i] \odot \mathbf{s}_i + \mathbf{D} \mathbf{H}_{\text{sort}}[i], \quad (13)$$

where $\mathbf{H}_{\text{sort}}[i] \in \mathbb{R}^d$ is the input feature of the i -th atom after nuclear-charge ordering; $\mathbf{W}_g \in \mathbb{R}^{s \times d}$ is the linear projection that computes the selection gate $\sigma(\cdot)$ is the element-wise SiLU / Swish activation; $\mathbf{g}_i \in \mathbb{R}^s$ is a input-dependent gating vector (s hidden states); $\mathbf{A} \in \mathbb{R}^{s \times s}$ is the shared transition matrix (trainable but not time-varying); $\tilde{\mathbf{A}}_i \in \mathbb{R}^{s \times s}$ is the gated version of $\mathbf{A}$ for token i ; $\mathbf{1} \in \mathbb{R}^s$ is all-ones column vector, so $\mathbf{g}_i \mathbf{1}^\top$ broadcasts gating to full matrix; $\odot$ — element-wise (Hadamard) product; $\mathbf{B}[i], \mathbf{C}[i] \in \mathbb{R}^s$ are input-dependent input / output vectors produced by linear projections of $\mathbf{H}_{\text{sort}}[i]$; $\mathbf{s}_i \in \mathbb{R}^s$ is the latent state (hidden memory) of the SSM at position i ; $\mathbf{D} \in \mathbb{R}^{d \times s}$ skip-connection matrix; and $\mathbf{H}_{\text{glob}}[i] \in \mathbb{R}^d$ the global contextual embedding for atom i after the inverse permutation.

Thanks to the associative scan, the entire sequence is processed in $\mathcal{O}(ns)$ time and $\mathcal{O}(ns)$ memory, independent of molecular size n . The selection mechanism dynamically filters distant yet relevant chemical contexts.

5.3.3 Topological augmentation

To compensate for the loss of explicit edge information during serialization, we concatenate each atom feature with its Random-Walk Structural Encoding before sorting, ensuring that ring membership and shortest-path distances remain recoverable by the SSM kernel.

By coupling chemically meaningful order with linear-complexity global context, NC-Mamba simultaneously alleviates over-squashing endemic to deep MPNNs, avoids

the quadratic bottleneck of graph transformers, and foregrounds heteroatoms and π -systems critical for property prediction without introducing any learnable sorting parameters.

5.4 KAN dynamic mixture module (KDM)

Simple concatenation or static attention weights treat "noise from distant alkyl chains" and "signal from ortho-nitro alerts" equally. KDM replaces this heuristic with a data-driven, non-linear gating mechanism that (i) learns how much local vs. global information each atom needs, and (ii) retains the universal approximation power of Kolmogorov–Arnold networks while using fewer parameters than a vanilla MLP of comparable width. KDM’s pipeline is described in Fig. 1(c). Local embeddings $\mathbf{H}_{\text{loc}}$ and global embeddings $\mathbf{H}_{\text{glob}}$ are first concatenated and refined by multi-head attention, then fed into a residual stack of KAN-based feed-forward blocks (ResKAN-Layer + KAN-FFN) for adaptive, non-linear fusion. The final output is the per-atom embedding $\mathbf{H}_v$ ready for graph-level readout.

(1) For every node v we first form the two-stream input

$$\mathbf{z}_v = [\mathbf{H}_{\text{loc}} \parallel \mathbf{H}_{\text{glob}}] \in \mathbb{R}^{2d} \quad (14)$$

A single-hidden-layer KAN spline block transforms $\mathbf{z}_v$:

$$\mathbf{u}_v = \text{KAN}_\theta(\mathbf{z}_v) = \sum_{j=1}^{2d} \phi_j(z_v^{(j)}) \in \mathbb{R}^s, \quad \phi_j(x) = \sum_{k=1}^K c_{jk} \cdot B_k(x; \Delta) \quad (15)$$

where $z_v^{(j)} \in \mathbb{R}$ the j -th coordinate of the concatenated vector $\mathbf{z}_v$; $\phi_j : \mathbb{R} \rightarrow \mathbb{R}$ learnable univariate function implemented by B-splines; $B_k(x; \Delta)$ the k -th B-spline basis of fixed grid spacing Δ (uniform knots, degree = 3); $c_{jk} \in \mathbb{R}$ the trainable coefficient for basis B_k on dimension j ; K the number of basis functions per dimension (hyper-parameter, typically $K = 5-8$); $\text{KAN}_\theta(\cdot)$ the Kolmogorov–Arnold outer sum: $\sum_{j=1}^{2d} \phi_j(z_v^{(j)})$, output dimension s (here $s = d$); $\mathbf{u}_v \in \mathbb{R}^s$ the KAN output that feeds the subsequent linear gate. The spline grid is shared across all nodes, so the module adds only $\mathcal{O}(2dK + sd)$ parameters independent of graph size.

(2) A linear gate computes atom-specific mixture weights

$$[\alpha_v; \beta_v] = \text{softmax}(\mathbf{W}\mathbf{u}_v) \in \mathbb{R}^2 \quad \alpha_v + \beta_v = 1. \quad (16)$$

where $\mathbf{W} \in \mathbb{R}^{2 \times s}$ the learnable linear map that compresses the s -dim KAN output $\mathbf{u}_v$ into a 2-dim logit vector; $[\alpha_v; \beta_v] \in \mathbb{R}^2$ the softmax-normalized weights controlling the local-vs-global contribution for atom v .

(3) The node embedding is produced by residual blending

$$\mathbf{H}_v = \alpha_v \odot \mathbf{H}_{\text{loc}} + \beta_v \odot \mathbf{H}_{\text{glob}} + \mathbf{H}_{\text{loc}} \quad (17)$$

where $\alpha_v, \beta_v \in \mathbb{R}$ the softmax-derived scalar weights for atom v ; $\odot$ the element-wise (Hadamard) product; since α_v is a scalar, this reduces to ordinary scaling; the residual skip connection of $\mathbf{H}_{\text{loc}}$ stabilizes gradient flow and accelerates convergence.

KDM yields adaptive, expressive and efficient fusion: per-atom gates α_v, β_v let hydroxyls foreground local polarity while aromatic nitrogens exploit global conjugation; KAN splines deliver higher approximation power per parameter than ReLU-MLPs, enhancing quality across varied molecular scales; a shared spline grid plus residual skip connection accelerates convergence and curbs over-fitting, as experimentally verified.

5.5 Prediction module

The prediction module employs attention pooling to aggregate node embeddings into graph-level representations, incorporates edge information through edge-aware mechanisms, and utilizes task-specific MLP heads for molecular property prediction, trained end-to-end through empirical risk minimization. The prediction pipeline is described in Fig. 1(d).

5.5.1 Graph-level readout with edge awareness

A lightweight attention pool aggregates node embeddings while considering edge connectivity. The edge context $\mathbf{m}_v^{\text{edge}}$ for each node v is derived from the MPNN-learned edge features $\mathbf{E}_{\text{loc}}$ through neighborhood aggregation:

$$\mathbf{m}_v^{\text{edge}} = \bigoplus_{u \in \mathcal{N}(v)} \text{MLP}_{\text{edge-context}} \left(\mathbf{e}_{uv}^{(T)} \right) \quad (18)$$

where $\mathbf{e}_{uv}^{(T)} \in \mathbf{E}_{\text{loc}}$ is the final edge representation from the MPNN module, $\bigoplus$ denotes mean-pooling, and $\text{MLP}_{\text{edge-context}}$ transforms the raw edge features into a contextual representation compatible with node embeddings.

The attention weights are then computed by incorporating both node features and their edge contexts:

$$\gamma_v = \frac{\exp \left(\mathbf{q}^\top \left[\mathbf{H}_v \parallel \mathbf{m}_v^{\text{edge}} \right] \right)}{\sum_u \exp \left(\mathbf{q}^\top \left[\mathbf{H}_u \parallel \mathbf{m}_u^{\text{edge}} \right] \right)}, \quad \mathbf{H}_{\mathcal{G}} = \sum_v \gamma_v \mathbf{H}_v \quad (19)$$

where the vector $\mathbf{q} \in \mathbb{R}^{2d}$ is learned jointly with downstream tasks, enabling the attention mechanism to consider both the node features and their local edge contexts during graph-level aggregation.

5.5.2 Edge feature integration

To explicitly leverage the bond-level information captured in $\mathbf{E}_{\text{loc}}$, we compute a global edge representation through direct pooling of the MPNN edge features:

$$\mathbf{H}_{\mathcal{G}}^{\text{edge}} = \frac{1}{|\mathcal{E}|} \sum_{(u,v) \in \mathcal{E}} \text{MLP}_{\text{global-edge}} \left(\mathbf{e}_{uv}^{(T)} \right) \in \mathbb{R}^{d_{\text{edge}}} \quad (20)$$

where $\mathbf{e}_{uv}^{(T)} \in \mathbf{E}_{\text{loc}}$ are the final edge embeddings from the MPNN module, and $\text{MLP}_{\text{global-edge}}$ projects these edge features to a compressed representation space.

The final graph-level embedding combines both node and edge information:

$$\mathbf{H}_{\mathcal{G}}^{\text{final}} = [\mathbf{H}_{\mathcal{G}} \parallel \mathbf{H}_{\mathcal{G}}^{\text{edge}}] \in \mathbb{R}^{d+d_{\text{edge}}} \quad (21)$$

5.5.3 Task-specific heads

For regression tasks (log-solubility, etc.) we use:

$$\hat{y} = \mathbf{w}_{\text{reg}}^{\top} \sigma(\mathbf{W}_{\text{reg}} \mathbf{H}_{\mathcal{G}}^{\text{final}} + \mathbf{b}_{\text{reg}}) \quad (22)$$

while for C -class classification (toxicity, etc.) we employ:

$$\hat{\mathbf{p}} = \text{softmax}(\mathbf{W}_{\text{clf}} \mathbf{H}_{\mathcal{G}}^{\text{final}} + \mathbf{b}_{\text{clf}}) \in \mathbb{R}^C \quad (23)$$

where $\mathbf{H}_{\mathcal{G}}^{\text{final}} \in \mathbb{R}^{d+d_{\text{edge}}}$ is the final graph-level embedding combining node and edge information, $\mathbf{W}_{\text{reg}} \in \mathbb{R}^{d' \times (d+d_{\text{edge}})}$, $\mathbf{b}_{\text{reg}} \in \mathbb{R}^{d'}$ are the hidden-layer parameters for regression (default $d' = 256$), $\sigma(\cdot)$ is the SiLU activation function, $\mathbf{w}_{\text{reg}} \in \mathbb{R}^{d'}$ is the output weight vector for regression, $\hat{y} \in \mathbb{R}$ is the predicted scalar property, $\mathbf{W}_{\text{clf}} \in \mathbb{R}^{C \times (d+d_{\text{edge}})}$, $\mathbf{b}_{\text{clf}} \in \mathbb{R}^C$ are the parameters for C -class classification, $\hat{\mathbf{p}} \in \mathbb{R}^C$ is the predicted class-probability vector.

5.5.4 Training objective

We minimize the standard empirical risk:

$$\mathcal{L} = \frac{1}{|\mathcal{D}|} \sum_{(\mathcal{G}, y) \in \mathcal{D}} \ell(\hat{y}, y) \quad (24)$$

where ℓ is MSE for regression or cross-entropy for classification.

The computational complexity of the enhanced prediction module remains $\mathcal{O}(n + m)$, linear in the number of atoms and edges, preserving the model’s scalability for large-batch training on million-molecule libraries. By incorporating edge features $\mathbf{E}_{\text{loc}}$ through both local edge-aware attention ($\mathbf{m}_v^{\text{edge}}$) and global edge pooling ($\mathbf{H}_{\mathcal{G}}^{\text{edge}}$), the prediction module captures comprehensive molecular structural information, leading to more accurate and chemically meaningful property predictions.

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